

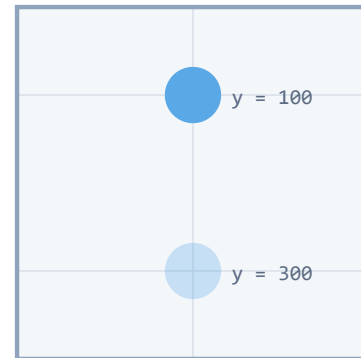
Making a variable

```
let y = 100;
```

`let` makes a new name. `y` is the name. `100` is what is in it.

Anywhere you could write a number, you can write the name instead.

```
circle(200, y, 60);
```



change the variable, not the shape

let goes at the top, outside `draw()`. Inside `draw()` it is created fresh sixty times a second, so it can never grow — your sketch will sit perfectly still and the cause will be four lines away from where you are looking.

Making it move

```
function draw() {
  background(24, 38, 60);
  circle(200, y, 60);
  y = y + 1;
}
```

Read it as “becomes”, not “equals”.

`y = y + 1` means: take what is in `y`, add one, put it back. In math that sentence has no solution. Here it is an instruction, not a claim.

Why one line moves it more than once

`draw()` is not run once. It runs about **60 times every second**, forever, until you stop the sketch.

So one `y = y + 1` is sixty pixels of movement per second.

Speed is just how much you add.

`y = y + 1` slow · `y = y + 6` fast

`y = y - 2` goes **up**, because `y` grows downward.

FRAME	Y
1	101
2	102
3	103
...	...
60	160

one second later

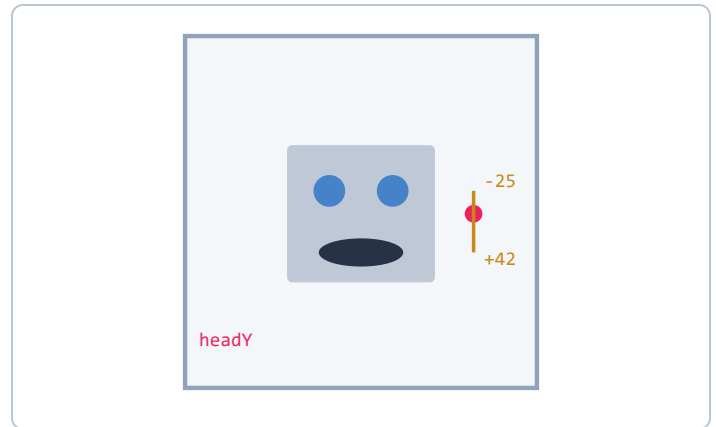
`background()` is at the top of `draw()` for the same reason. Without it, every frame paints on top of the last one and a moving shape leaves a trail.

Moving more than one shape

A drawing is many shapes at many heights. If they all move together, each one has to keep its **distance** from the others.

Pick **one point**, put it in a variable, and write every other shape as an offset from it.

```
rect(200, headY, 160, 150);  
circle(165, headY - 25, 34);  
ellipse(200, headY + 42, 90, 30);
```



The eyes are not at 165. They are 25 above *wherever the head is*. Leave one shape on a fixed number and it stays behind while everything else moves — the head slides away and the face floats in mid-air.

Variables you did not make

NAME	WHAT IS IN IT
width	the canvas width, set by <code>createCanvas</code>
height	the canvas height
mouseX	where the cursor is across, right now
mouseY	where the cursor is down, right now

p5 updates `mouseX` and `mouseY` every frame. Some variables you change; some the system changes for you. Both are just names for numbers.

On the AP exam

What you write

```
let y = 100;  
y = y + 1;
```

Exam Reference Sheet

```
y ← 100  
y ← y + 1
```

The AP exam is not written in JavaScript. It uses College Board's own notation, and the arrow `←` means **becomes** — the same word we use for `=`. No `let`, no semicolon.

Nothing you know yet can stop a falling shape. The variable keeps growing and the shape keeps going, off the screen and past it. To stop it, the program has to ask a question — *has it gone too far?* That is next lesson.